



NORTHERN BEACHES SECONDARY COLLEGE

MANLY SELECTIVE CAMPUS

HIGHER SCHOOL CERTIFICATE

TRIAL EXAMINATION

2015

Mathematics Extension II

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Write your Student Number at the top of each page
- Answer Section I- Multiple Choice on Answer Sheet provided
- Answer Section II – Free Response in a separate booklet for each question.
- Board approved calculators and templates may be used.

Section I Multiple Choice

- 10 marks
- Attempt all questions
- Allow about 15 minutes for this section

Section II – Free Response

- 90 marks
- Each question is of equal value
- All necessary working should be shown in every question.
- Allow about 2 hours 45 minutes for this section

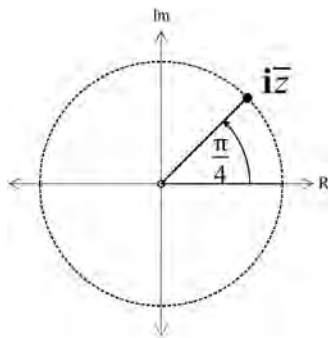
Weighting: 40%

Section 1: Multiple Choice (10 marks)

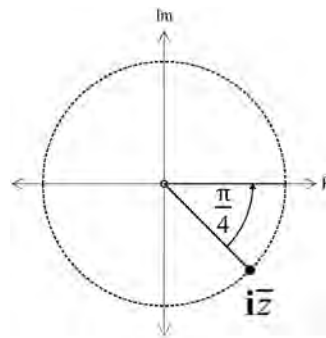
Indicate your answer on answer sheet provided.

Q1. Given the complex number z has $\text{Arg } z = \frac{\pi}{4}$, which of the following is a correct representation of $i\bar{z}$?

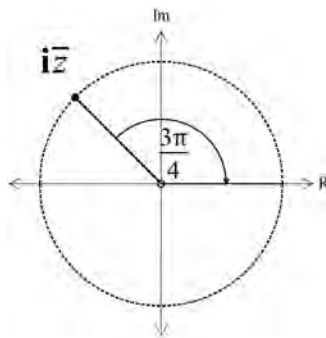
A



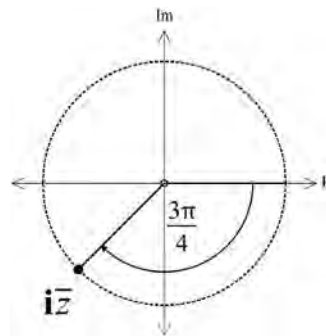
B



C



D



Q2. What is the eccentricity of the hyperbola $4x^2 - 9y^2 = 36$?

A $\frac{\sqrt{13}}{2}$

B $\frac{\sqrt{13}}{3}$

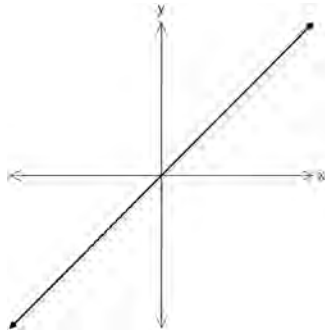
C $\frac{13}{4}$

D $\frac{13}{9}$

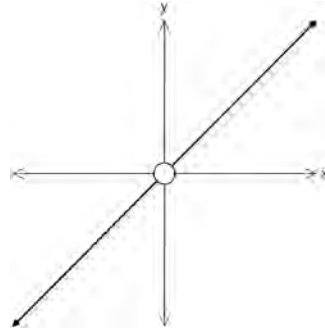
- Q3. The equation $\frac{x}{y} + \frac{y}{x} = 2$
defines y implicitly as a function of x .

Which of the following graphs best represents this implicit function?

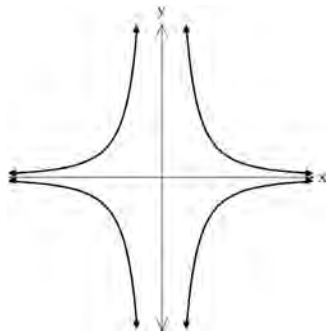
A



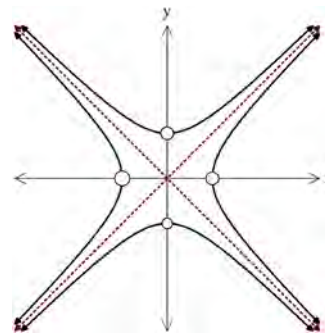
B



C



D



- Q4. The polynomial equation $x^3 + 3x^2 - 2x + 6 = 0$ has roots α , β and γ .

Which polynomial has roots $\alpha - 1$, $\beta - 1$, $\gamma - 1$?

- A $x^3 - 5x + 10 = 0$
 B $x^3 + 3x^2 - 2x + 14 = 0$
 C $x^3 + 6x^2 + x + 8 = 0$
 D $x^3 + 6x^2 + 7x + 8 = 0$

Q5. $\int x \sin 2x \, dx =$

A $-\frac{x}{2} \cos 2x + \frac{1}{4} \sin 2x + C$

B $-\frac{x}{2} \cos 2x - \frac{1}{4} \sin 2x + C$

C $\frac{x}{2} \cos 2x - \frac{1}{4} \sin 2x + C$

D $-2x \cos 2x + \sin 2x + C$

Q6. If $\int_1^4 f(x) \, dx = 6$, what is the value of $\int_1^4 f(5-x) \, dx = ?$

A 6

B 3

C 0

D -6

Q7. The point $P(z)$ moves on the complex plane according to the condition $|z - i| + |z + i| = 4$. The Cartesian equation of the locus of P is:

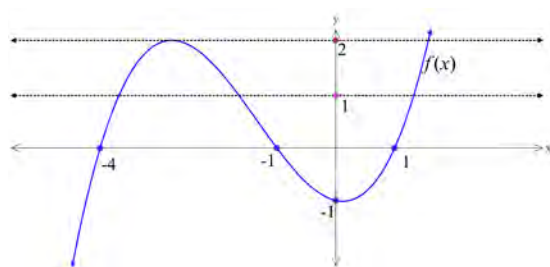
A $\frac{x^2}{4} + \frac{y^2}{3} = 1$

B $\frac{x^2}{3} + \frac{y^2}{4} = 1$

C $x^2 + y^2 = 1$

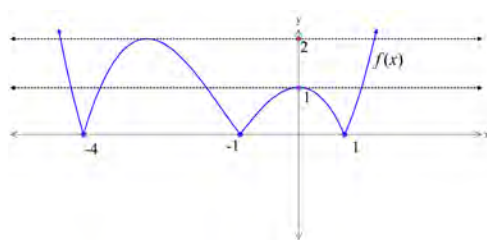
D $x^2 + y^2 = 4$

Q8.

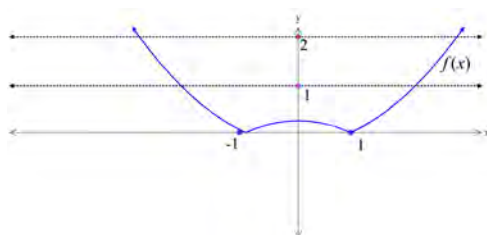


The graph $y = f(x)$ is shown in the figure above. Which of the following could be the graph of $y = f(|x|)$

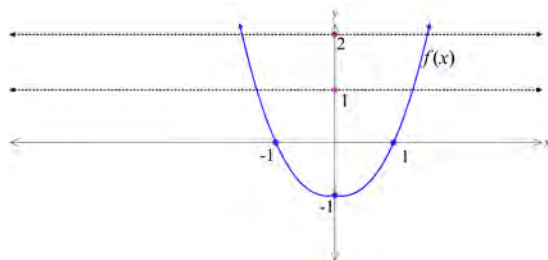
A



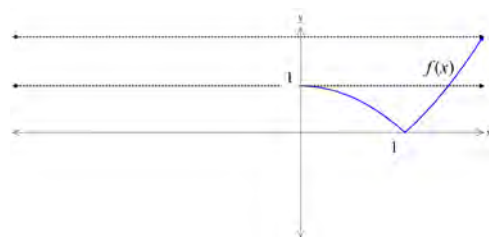
B



C



D



- Q9. Let R be the region between the graphs of $y = 1$ and $y = \sin x$ from $x = 0$ to $x = \frac{\pi}{2}$.
The volume of the solid obtained by revolving R around the x – axis is given by

- A $2\pi \int_0^{\frac{\pi}{2}} x \sin x \, dx$
- B $2\pi \int_0^{\frac{\pi}{2}} x \cos x \, dx$
- C $\pi \int_0^{\frac{\pi}{2}} \cos^2 x \, dx$
- D $\pi \int_0^{\frac{\pi}{2}} (1 - \sin x)^2 \, dx$

- Q10. What is the area of the largest rectangle that can be inscribed in the ellipse $4x^2 + 9y^2 = 36$?

- A $24\sqrt{2}$
- B $6\sqrt{2}$
- C 24
- D 12

End of Multiple Choice

Section II Total Marks is 90**Attempt Questions 11 – 16.****Allow approximately 2 hours & 45 minutes for this section.**

Answer all questions, starting each new question in a new booklet with your **student ID number** in the top right hand corner and the question number on the left hand side of your paper. All necessary working must be shown in each and every question.

Question 11. – Start New Booklet**15 marks**

a) Find $\sqrt{8 - 6i}$ (2)

b) Factorise $z^2 + 2iz + 15$ (2)

c) Given the expression $\frac{6x^2 + 3x + 1}{(x + 1)(x^2 + 1)}$

(i) Find numbers A , B and C such that

$$\frac{6x^2 + 3x + 1}{(x + 1)(x^2 + 1)} = \frac{A}{x + 1} + \frac{Bx + C}{x^2 + 1} \quad (2)$$

(ii) Find $\int \frac{6x^2 + 3x + 1}{(x + 1)(x^2 + 1)} dx$ (3)

d) Evaluate $\int_0^{\sqrt{3}} 75x^3 \sqrt{1 + x^2} dx$ (3)

e) Given the polynomial $P(x) = x^4 - x^3 + 7x^2 - 9x - 18$

(i) Show that $(x + 1)$ is a factor of $P(x)$ (1)

(ii) Factorise $P(x)$ over the complex field. (2)

Question 12 Start New Booklet**15 Marks**

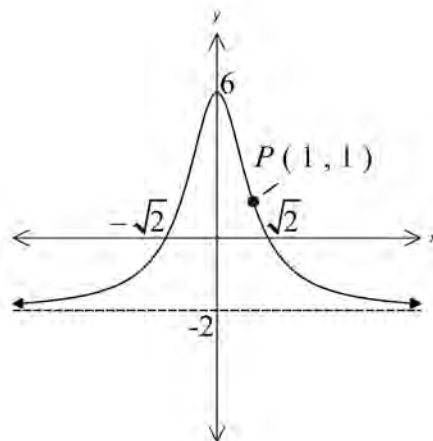
- a) Find the exact value of $\int_{-\sqrt{2}}^{\sqrt{2}} \frac{e^x}{e^x + 1} dx$. (3)

(Give your answer in simplest exact form.)

- b) The point P on the Argand diagram represents the complex number z , where z satisfies:

$$\frac{2}{z} + \frac{2}{\bar{z}} = 1$$

- (i) Find the Cartesian equation of the locus of P as z varies. (2)
- (ii) Sketch the locus of P (1)
- c) The diagram below shows the graph $y = f(x)$. The point P has the coordinates $(1, 1)$.

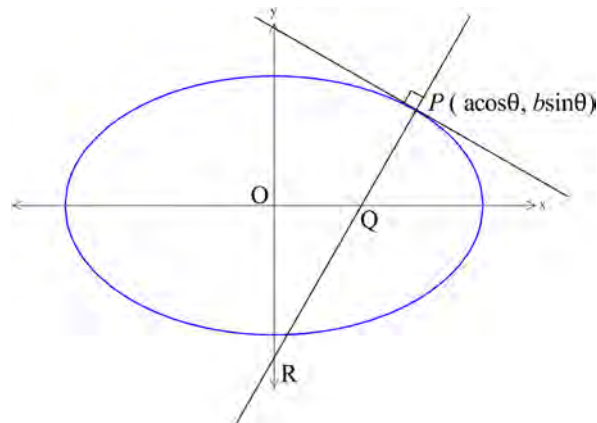


On separate diagrams sketch the graphs of the following functions.

Each sketch must be at least $\frac{1}{3}$ of a page and must be clearly labelled, showing the coordinate axes, origin and all significant features.

- (i) $y = |f(x)|$ (1)
- (ii) $y = \log\{f(x)\}$ (1)
- (iii) $y = [f(x)]^2$ (2)
- (iv) The inverse function $y = f^{-1}(x)$ (2)
- (v) $y = \frac{1}{6 - f(x)}$ (3)

a)



$P (a\cos\theta, b\sin\theta)$, where $0 < \theta < \frac{\pi}{2}$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

- (i) Show that the equation of the normal at P is

$$a(\sin\theta)x - (b\cos\theta)y = (a^2 - b^2)\sin\theta\cos\theta \quad (2)$$

- (ii) Show that ΔQOR has an area given by

$$A = \frac{(a^2 - b^2)^2}{2ab} \sin\theta\cos\theta \quad (2)$$

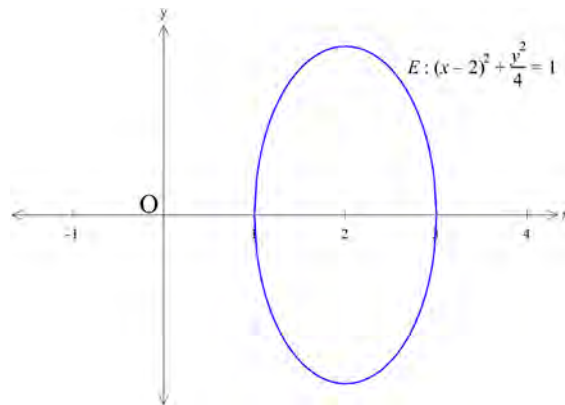
- (iii) Find the maximum value of A (2)

- (iv) Find the coordinates of P for which the maximum value of A occurs. (1)

Question 13 continues next page.

Question 13 continued.

b)



The region enclosed by the ellipse $E : (x - 2)^2 + \frac{y^2}{4} = 1$ is rotated through one complete revolution around the y axis.

- (i) Use the method of cylindrical shells to show that the volume V of the solid of revolution is given by

$$V = 8\pi \int_1^3 x \sqrt{1 - (x - 2)^2} \, dx \quad (2)$$

- (ii) Hence find the volume of the solid of revolution in its simplest exact form. (4)

- c) Find the gradient of the tangent to curve $x^2 + xy - y^2 = 11$ at the point $P(3,1)$ (2)

Question 14 – Start New Booklet

15 Marks

a) Show that $\int \frac{\sin^3 x}{\cos^4 x} dx = \frac{1}{3} \sec^3 x - \sec x$ (3)

b) (i) Show that $\int_{-a}^0 f(x) dx = \int_0^a f(-x) dx$ (2)

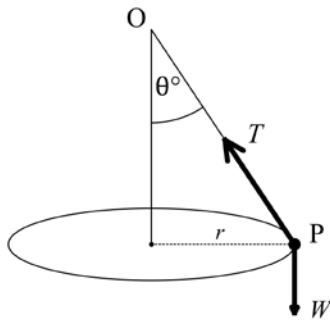
(ii) Deduce that $\int_{-a}^a f(x) dx = \int_0^a \{f(x) + f(-x)\} dx$ (1)

(iii) Hence, or otherwise, evaluate $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \frac{1}{1 + \sin x} dx$ (3)

(Express your answer in simplest exact form)

- c) A body P of mass 0.5kg is suspended from a fixed point O by means of a light rod of length 1 metre.

The mass is rotated in a horizontal circle at a constant speed $v \text{ ms}^{-1}$. The rod makes an angle θ with the downward vertical direction as shown in the diagram below.



The tension in the rod is T newtons and the weight of P is W newtons. The radius of the circle is r metres.

Assume $g = 9.8 \text{ ms}^{-2}$ and $\theta = 30^\circ$

- (i) Show that $\tan \theta = \frac{v^2}{rg}$ (3)
- (ii) Find the tension T . (1)
- (iii) Find the speed $v \text{ ms}^{-1}$ of P (1)
- (iv) Find the period of the motion. (1)

Question 15 – Start a new booklet

- a) (i) By considering $z^9 - 1$ as the difference of two cubes, show that

$$z^9 - 1 = (z - 1)(z^2 + z + 1)(z^6 + z^3 + 1) \quad (1)$$

- (ii) Use the result $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$ to show that

$$z^8 + z^7 + z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = (z^2 + z + 1)(z^6 + z^3 + 1) \quad (1)$$

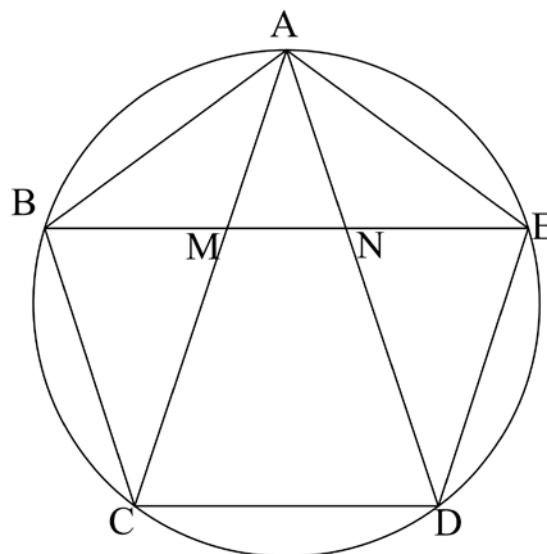
- (iii) Solve $z^9 - 1 = 0$ and hence determine the six solutions of $z^6 + z^3 + 1 = 0$

(2)

- (iv) Hence show that $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = \cos \frac{\pi}{9}$

(2)

b)



$ABCDE$, where $AB = AE$, is a pentagon inscribed in a circle. BE intersects AC at M and AD at N respectively.

- (i) Show that $\angle BEA = \angle ACE$ (2)

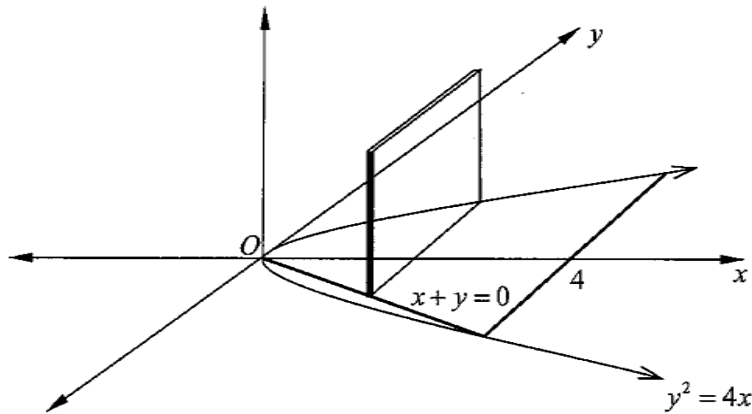
- (ii) Hence show that $CDNM$ is a cyclic quadrilateral (3)

Question 15 – continues on next page.

Question 15 – continued

- c) The base of a solid is the region bounded by the curve $y^2 = 4x$ and the lines $x + y = 0$ and $x = 4$.

Every cross section perpendicular to the x axis is a square having a side with one end point on the line $x + y = 0$ and the other on the curve $y^2 = 4x$



- (i) Show that the area A of the cross section is given by

$$A = 4x + x^2 + 4x^{\frac{3}{2}} \quad (2)$$

- (ii) Hence find the volume of the solid. (2)

Question 16 – Start a New Booklet

15 Marks

a) Let $I_n = \int_0^1 (1-x^2)^n dx$ for $n \geq 1$

(i) Show that $I_n = \frac{2n}{2n+1} \cdot I_{n-1}$ (3)

(ii) Evaluate I_3 (1)

b) Consider the rectangular hyperbola $xy = c^2$, where $c > 0$

(i) P and Q are points on the hyperbola with coordinates $(cp, \frac{c}{p})$ and $(cq, \frac{c}{q})$ respectively.

Prove that the equation of the chord joining P and Q is given by:

$$x + pqy = c(p + q) \quad (2)$$

(ii) The chord PQ intersects the x and y axes at M and respectively N respectively.

Prove that $PN = QM$ (3)

c) A polynomial $P(x)$ is divided by $x^2 - a^2$ where $a \neq 0$, and the remainder is $px + q$.

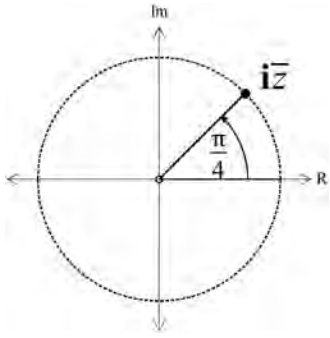
(i) Show that

$$p = \frac{1}{2a} [P(a) - P(-a)] \text{ and}$$

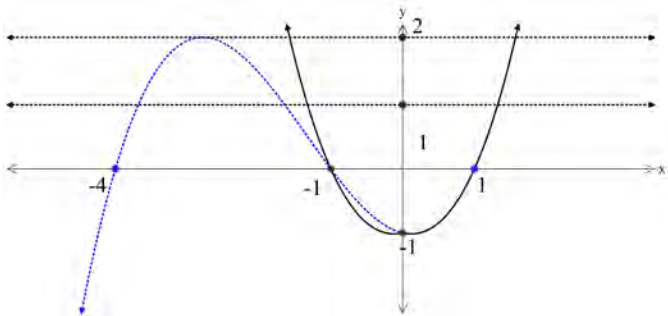
$$q = \frac{1}{2} [P(a) + P(-a)] \quad (3)$$

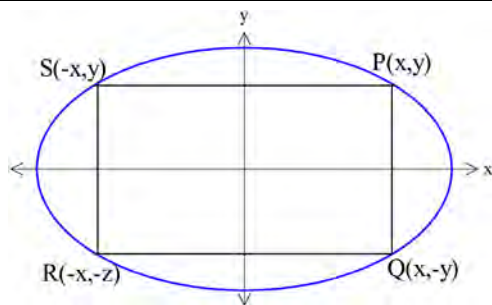
(ii) Find the remainder when $P(x) = x^n - a^n$ for n a positive integer, is divided by $x^2 - a^2$. (3)

End of Examination

Q1	 Conjugate of z followed by 90 degree anticlockwise rotation.	A
Q2	$4x^2 - 9y^2 = 36$ $\frac{x^2}{9} - \frac{y^2}{4} = 1 \Rightarrow a^2 = 9 \quad b^2 = 4$ $e^2 = 1 + \frac{b^2}{a^2}$ $e^2 = \frac{13}{9}$ $e = \sqrt{\frac{13}{9}} = \frac{\sqrt{13}}{3}$	B
Q3	$\frac{x}{y} + \frac{y}{x} = 2$ $x^2 + y^2 = 2xy$ $x^2 - 2xy + y^2 = 0$ $(x - y)^2 = 0$ $y = x; x \neq 0 \quad y \neq 0$	B
Q4	$x^3 + 3x^2 - 2x + 6 = 0$ $\alpha - 1; \beta - 1; \gamma - 1$ $\Rightarrow \begin{aligned} & y = x - 1 \\ & x = y + 1 \end{aligned}$ $(y + 1)^3 + 3(y + 1)^2 - 2(y + 1) + 6 = 0$ $y^3 + 3y^2 + 3y + 1 + 3y^2 + 6y + 3 - 2y - 2 + 6 = 0$ $y^3 + 6y^2 + 7y + 5 = 0$ $\Rightarrow x^3 + 6x^2 + 7x + 5 = 0$	D

Q5	$\int x \sin 2x \, dx = -\frac{1}{2} x \cos 2x - \int -\frac{1}{2} (\cos 2x) \, dx$ $u = x \Rightarrow du = 1$ $dv = \sin 2x \Rightarrow v = -\frac{1}{2} \cos 2x$ $\int x \sin 2x \, dx = -\frac{1}{2} x \cos 2x + \frac{1}{2} \frac{1}{2} \sin 2x$ $= -\frac{x}{2} \cos 2x + \frac{1}{4} \sin 2x$	A
Q6	$\int_1^4 f(5-x) \, dx$ $u = 5-x$ $x = 1 \Rightarrow u = 4$ $x = 4 \Rightarrow u = 1$ $dx = -du$ $\int_4^1 f(u) (-du)$ $= -\int_4^1 f(u) \, du$ $= \int_1^4 f(u) \, du$ $= \int_1^4 f(x) \, dx = 6$	A
Q7	$ \bar{z} - i + z + i = 4$ "From $PS + PS' = 2b$" the locus is an ellipse with foci $(0, \pm i)$ $\therefore 2b = 4$ $\Rightarrow b^2 = 4$ $\therefore \frac{x^2}{3} + \frac{y^2}{4} = 1$	B

Q8		C
Q9	Using disc method $\begin{aligned}\delta V &= \pi(R^2 - r^2)\delta x \\ &= \pi[1^2 - (\sin x)^2]\delta x \\ &= \pi(\cos^2 x)\delta x\end{aligned}$ $V = \pi \int_0^{\frac{\pi}{2}} \cos^2 x \, dx$	C



$$4x^2 + 9y^2 = 36$$

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$Area = (2x)(2y) = 4xy$$

$$A^2 = 16x^2y^2$$

$$\frac{A^2}{4} = 16x^2\left(\frac{y^2}{4}\right)$$

$$= 16x^2\left(1 - \frac{x^2}{9}\right)$$

$$H = \frac{A^2}{4} = 16x^2 - \frac{16x^4}{9}$$

$$\frac{dH}{dx} = 32x - \frac{64x^3}{9} = 0$$

$$\Rightarrow 32 - \frac{64x^2}{9} = 0$$

$$x^2 = \frac{9}{2}$$

$$\Rightarrow y^2 = 2$$

$$A^2 = 16\left(\frac{9}{2}\right)^2 = 144$$

$$A = 12$$

Q10

D

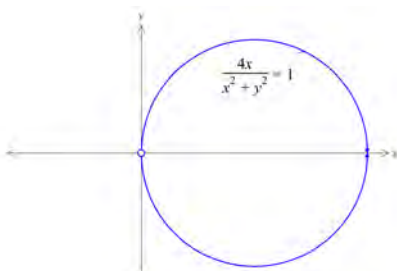
Question 11

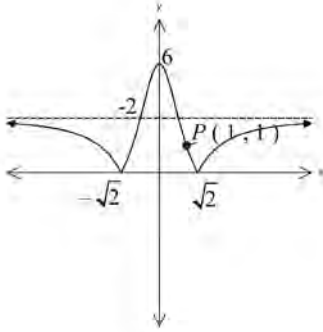
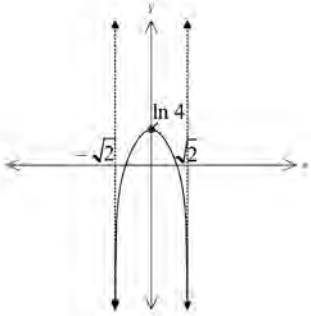
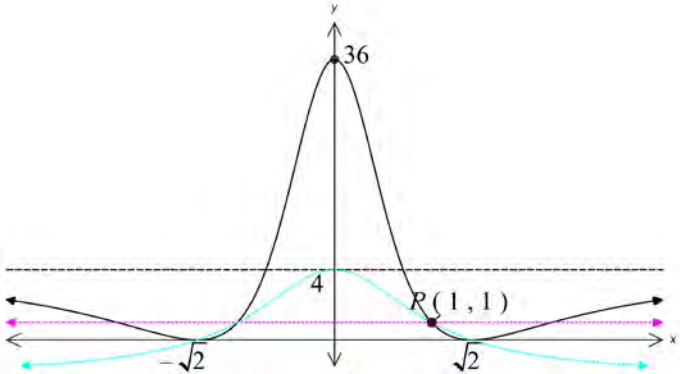
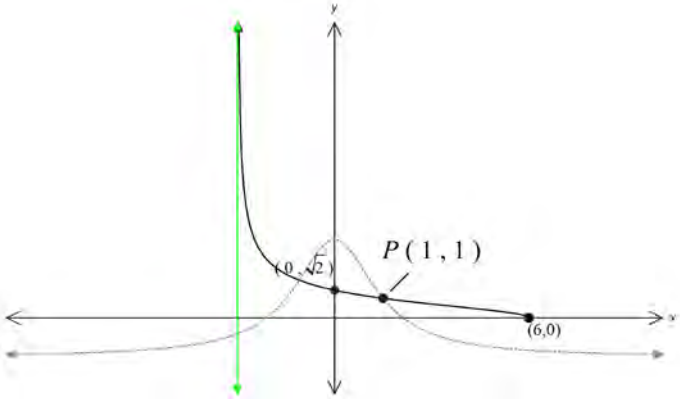
a	$x + iy = \sqrt{8 - 6i}$ $(x + iy)^2 = 8 - 6i$ $x^2 - y^2 = 8 \quad xy = -3 \Rightarrow y = -\frac{3}{x}$ $\therefore x^2 - \frac{9}{x^2} = 8$ $x^4 - 8x^2 - 9 = 0$ $(x^2 - 9)(x^2 + 1) = 0$ $x^2 - 9 = 0 \quad x^2 + 1 = 0$ $x = \pm 3 \quad \text{No solution}$ $y = \frac{3}{x} \Rightarrow y = \pm 1$ $\sqrt{8 - 6i} = \pm(3 - i)$	2 marks – correct solution 1 mark – obtains equations $x^2 - y^2 = 8$ $xy = -3$
b	$z^2 + 2iz + 15 = 0$ $(z + 5i)(z - 3i) = 0$	2 marks – correct solution 1 mark – solves equation to obtain $z = -5i, 3i$
c-i	$\frac{6x^2 + 3x + 1}{(x + 1)(x^2 + 1)} = \frac{A}{x + 1} + \frac{Bx + C}{x^2 + 1}$ $(6x^2 + 3x + 1) = A(x^2 + 1) + (Bx + C)(x + 1)$ $= (A + B)x^2 + (B + C)x + A + C$ $A + B = 6 \quad \textcircled{1}$ $B + C = 3 \quad \textcircled{2}$ $A + C = 1 \quad \textcircled{3}$ $\textcircled{3} - \textcircled{2} \quad A - B = -2 \quad \textcircled{4}$ $\textcircled{1} + \textcircled{4} \quad 2A = 4$ $\Rightarrow A = 2$ $B = 4$ $C = -1$	2 marks – correct solution 1 mark – attempts to equate coefficients from LHS and RHS

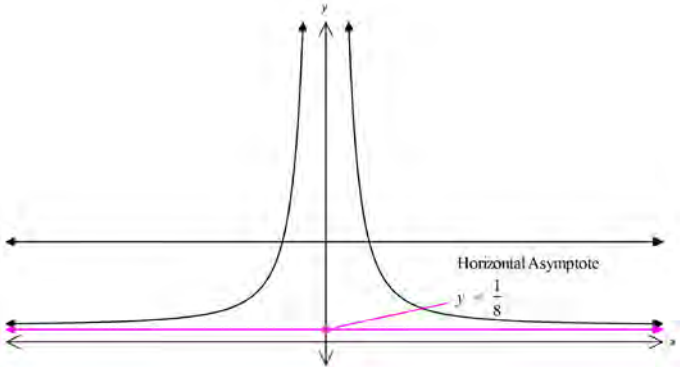
c-ii	$\int \frac{6x^2 + 3x + 1}{(x+1)(x^2+1)} dx$ $= \int \left\{ \frac{2}{x+1} + \frac{4x-1}{x^2+1} \right\} dx$ $= \int \frac{2}{x+1} dx + \int \frac{4x-1}{x^2+1} dx$ $= \int \frac{2}{x+1} dx + \int \frac{4x}{x^2+1} dx - \int \frac{1}{x^2+1} dx$ $= 2\ln(x+1) + 2\ln(x^2+1) - \tan^{-1} x + K$	3 marks – correct 2 marks – two correct terms from values of A, B and C 1 mark – obtains one correct term from values of A, B and C
d	Option 1 $\int_0^{\sqrt{3}} 75x^3 \sqrt{1+x^2} dx$ $u = 1 + x^2 \Rightarrow du = 2x dx$ $dx = \frac{du}{2x}$ $x^2 = u - 1$ $\Rightarrow x = 0 \quad u = 1$ $x = \sqrt{3} \quad u = 4$ $\int_1^4 75x^3 \sqrt{u} \frac{du}{2x} = \frac{75}{2} \int_1^4 x^2 \sqrt{u} du$ $= \frac{75}{2} \int_1^4 (u-1) \sqrt{u} du$ $= \frac{75}{2} \int_1^4 u^{\frac{3}{2}} - u^{\frac{1}{2}} du$ $= \frac{75}{2} \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right]_1^4$ $= \frac{75}{2} \left[\left(\frac{2}{5} 4^{\frac{5}{2}} - \frac{2}{3} 4^{\frac{3}{2}} \right) - \left(\frac{2}{5} \times 1 - \frac{2}{3} \times 1 \right) \right]$ $= 290$	3 marks – correct solution 2 marks – correct transformation of integrand and limits of integration 1 mark – correct transformation of integrand OR limits of integration NOTE: Several alternate techniques available to solve this question. Solution using $x = \tan \theta$ not included here.

d	Option 2 $\int_0^{\sqrt{3}} 75x^3 \sqrt{1+x^2} \, dx$ $= 75 \int_0^{\sqrt{3}} x \times x^2 \sqrt{1+x^2} \, dx$ $= 75 \int_0^{\sqrt{3}} x \times [(1+x^2) - 1] \sqrt{1+x^2} \, dx$ $= \frac{75}{2} \left\{ \int_0^{\sqrt{3}} 2x \times (1+x^2)^{\frac{3}{2}} - 2x(1+x^2)^{\frac{1}{2}} \, dx \right\}$ $= \frac{75}{2} \left[\frac{2}{5}(1+x^2)^{\frac{5}{2}} - \frac{2}{3}(1+x^2)^{\frac{3}{2}} \right]_0^{\sqrt{3}}$ $= \text{see above}$	
e-i	$P(x) = x^4 - x^3 + 7x^2 - 9x - 18$ $P(-1) = (-1)^4 - (-1)^3 + 7(-1)^2 - 9(-1) - 18 = 0$ $\therefore (x + 1)$ is a factor	1 mark – correct substitution and evaluation
e-ii	$P(2) = 0 \therefore (x - 2) \text{ is a factor}$ $\therefore (x + 1)(x - 2) = (x^2 - x - 2) \text{ is a factor}$ By long division or comparing coefficients $P(x) = (x^2 - x - 2)(x^2 + 9)$ $= (x + 1)(x - 2)(x + 3i)(x - 3i)$	2 marks – correct solution 1 mark – determines (x-2) is a factor

Q12

a	$\int_{-\sqrt{2}}^{\sqrt{2}} \frac{e^x}{e^x + 1} dx$ $= [\ln(e^x + 1)]_{-\sqrt{2}}^{\sqrt{2}}$ $= \ln\left(\frac{e^{\sqrt{2}} + 1}{e^{-\sqrt{2}} + 1}\right)$ $= \ln\left(\frac{e^{\sqrt{2}} + 1}{1 + e^{\sqrt{2}}}\right)$ $= \ln(e^{\sqrt{2}})$ $= \sqrt{2}$	3 marks – correct solution 2 marks – correct primitive function and simplification to $\ln\left\{\frac{(e^{\sqrt{2}}) + 1}{e^{-\sqrt{2}} + 1}\right\}$ 1 mark – correct primitive function
b-i	$\frac{2}{z} + \frac{2}{z} = 1$ $z = x + iy \Rightarrow \bar{z} = x - iy$ $\frac{2}{x + iy} + 2(x - iy) = 1$ $\frac{2x + 2iy + 2x - 2iy}{x^2 + y^2} = 1$ $4x = x^2 + y^2$ $x^2 - 4x + y^2 = 0$ $(x - 2)^2 + y^2 = 4$ Circle with centre (2,0) and radius 2 except (0,0) not included.	2 marks – correct solution origin need not be excluded 1 mark – obtains $x^2 + y^2 = 4x$ or equivalent expression.
b-ii		1 mark – correct solution – origin must be excluded.

c-i		1 mark – correct solution
c-ii		1 mark – correct solution
c-iii		2 marks – correct solution (must show horizontal asymptote and y intercept) 1 mark – correct asymptote or y intercept.
c-iv		2 marks – correct solution (Must have x, y axes and vertical asymptote) 1 mark attempts to reflect $y = f(x)$ in $y = x$ with a correct intercept or vertical asymptote.

C-V		3 marks – correct solution (must show correct horizontal asymptote) 2 marks – correct shape with a horizontal asymptote 1 mark – correct shape.
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Q13

a-i	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\frac{d}{dx} \left\{ \frac{x^2}{a^2} + \frac{y^2}{b^2} \right\} = \frac{d}{dx} \{1\}$ $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{-b^2 x}{a^2 y}$ At $P(a \cos \theta, b \sin \theta)$ $m_1 = \frac{-b^2 a \cos \theta}{a^2 b \sin \theta} = \frac{-b \cos \theta}{a \sin \theta}$ For Normal $m_2 = \frac{a \sin \theta}{b \cos \theta}$ $y - y_1 = \frac{a \sin \theta}{b \cos \theta} (x - x_1)$ $y b \cos \theta - b^2 \sin \theta \cos \theta = x a \sin \theta - a^2 \sin \theta \cos \theta$ $(a \sin \theta)x - (b \cos \theta)y = (a^2 - b^2) \sin \theta \cos \theta$	2 marks correct solution 1 mark – obtains a correct expression for dy/dx.
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a-ii	At Q $y = 0$ $\Rightarrow (a \sin \theta)x = (a^2 - b^2) \sin \theta \cos \theta$ $x = \frac{(a^2 - b^2) \cos \theta}{a}$ At R $x = 0$ $\Rightarrow (-b \cos \theta)y = (a^2 - b^2) \sin \theta \cos \theta$ $y = \frac{-(a^2 - b^2) \sin \theta}{b}$ Area = $\frac{1}{2} (OQ)(OR) \quad (OR > 0)$ $= \frac{1}{2} \frac{(a^2 - b^2) \cos \theta}{a} \times \frac{(a^2 - b^2) \sin \theta}{b}$ $= \frac{(a^2 - b^2)^2 \sin \theta \cos \theta}{2ab}$	2 marks – correct solution 1 mark – obtains a correct expression for either x or y intercept.
a-iii	$\text{Area} = \frac{(a^2 - b^2)^2 \sin \theta \cos \theta}{2ab}$ $= \frac{(a^2 - b^2)^2 \sin 2\theta}{4ab}$ Maximum value of $\sin 2\theta = 1$ Max. Value of Area $\Rightarrow$ $= \frac{(a^2 - b^2)^2}{4ab}$	2 marks – correct solution 1 mark – transforms $\frac{1}{2} \sin \theta \cos \theta$ to $\sin(2\theta)$ - obtains a correct expression for $\frac{dA}{d\theta}$
a-iv	Max Value of $\sin 2\theta$ occurs at $2\theta = \pi/2$ $\Rightarrow \theta = \frac{\pi}{4}$ $P(a \cos \theta, b \sin \theta) = P\left(a \cos \frac{\pi}{4}, b \sin \frac{\pi}{4}\right)$ $= P\left(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}}\right)$	1 mark – correct solution

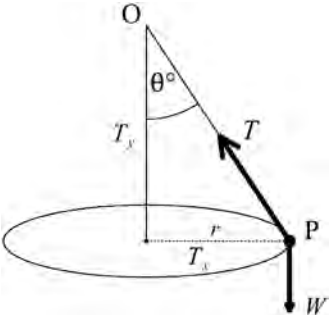
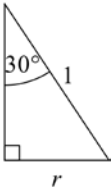
b-i	$\begin{aligned}\delta V &= \text{Circumference} \times \text{Height} \times \text{Thickness} \\ &= 2\pi x \times 2y \times \delta x \\ &= 4\pi xy \delta x\end{aligned}$ $\begin{aligned}V &= \lim_{\delta x \rightarrow 0} \sum_1^3 4\pi xy \delta x \\ &= \int_1^3 4\pi xy \, dx\end{aligned}$ $\begin{aligned}(x-2)^2 + \frac{y^2}{4} &= 1 \\ y^2 &= 4[1 - (x-2)^2] \\ y &= 2\sqrt{1 - (x-2)^2}\end{aligned}$ $\begin{aligned}V &= \int_1^3 4\pi x \cdot 2\sqrt{1 - (x-2)^2} \, dx \\ &= 8\pi \int_1^3 x \sqrt{1 - (x-2)^2} \, dx\end{aligned}$	2 marks – correct solution 1 mark – forms a correct expression for δV
b-ii	Let $u = x - 2 \Rightarrow dx = du$ $\therefore x = u + 2$ $x = 1 \Rightarrow u = -1$ $x = 3 \Rightarrow u = 1$ $\begin{aligned}V &= 8\pi \int_{-1}^1 (u+2)\sqrt{1-u^2} \, du \\ &= 8\pi \int_{-1}^1 u\sqrt{1-u^2} \, du + 16\pi \int_{-1}^1 \sqrt{1-u^2} \, du\end{aligned}$ But $f(u) = u\sqrt{1-u^2}$ $f(-u) = (-u)\sqrt{1-(-u)^2} = -(u\sqrt{1-u^2})$ $\therefore f(u) = -f(-u) \therefore$ odd $\therefore \int_{-a}^a \text{odd function} = 0$ $\begin{aligned}\therefore V &= 8\pi \times 0 + 16\pi \int_{-1}^1 \sqrt{1-u^2} \, du \\ &= 16\pi \times \text{Area of semicircle with radius } 1 \\ &= 16\pi \times \frac{1}{2} 1^2 \\ &= 8\pi\end{aligned}$	4 marks – correct solution 3 marks – forms a correct integral expression and evaluates one term correctly. 2 marks – forms correct integral expression with correct limits 1 mark – changes limits correctly Note: Trig substitution solution also accepted – not shown here.

c	$x^2 + xy - y^2 = 11$ $\frac{d}{dx}\{x^2 + xy - y^2\} = \frac{d(11)}{dx}$ $2x + \frac{xdy}{dx} + \frac{ydx}{dx} - \frac{2ydy}{dx} = 0$ $x \frac{dy}{dx} - 2y \frac{dy}{dx} = -2x - y$ $\frac{dy}{dx}(x - 2y) = -2x - y$ $\frac{dy}{dx} = \frac{2x + y}{2y - x}$ At $P(3,1)$ $m = \frac{2 \times 3 + 1}{2 \times 1 - 3}$ $= -\frac{7}{1}$	2 marks – correct solution 1 mark – attempts to use product rule to find dy/dx

Question 14

a	$\int \frac{\sin^3 x}{\cos^4 x} dx = \frac{1}{3} \sec^3 x - \sec x$ $u = \cos x$ $du = -\sin x dx$ $dx = \frac{du}{-\sin x}$ $I = \int \frac{\sin^3 x}{u^4} - \frac{du}{\sin x}$ $= \int \frac{\sin^2 x}{u^4} du$ $= \int \frac{1 - \cos^2 x}{u^4} du$ $= \int \frac{1 - u^2}{u^4} du$ $= \int u^{-4} - u^{-2} du$ $= \frac{u^{-1}}{-1} - \frac{u^{-3}}{-3}$ $= -\frac{1}{u} + \frac{1}{3u^3}$ $= -\frac{1}{\cos x} + \frac{1}{3\cos^3 x}$ $= \frac{1}{3} \sec^3 x - \sec x$	3 marks – correct solution 2 marks – forms correct transformed integral 1 mark – substitutes for $\cos x$ and obtains $\int \frac{\sin^2 x}{u^4} du$
b-i	$\int_{-a}^0 f(x) dx$ $x = -u \Rightarrow dx = -du$ $x = -a \Rightarrow u = a$ $x = 0 \Rightarrow u = 0$ $\int_{-a}^0 f(x) dx = \int_a^0 f(-u)(-du)$ $= -\int_a^0 f(-u) du$ $= \int_0^a f(-u) du$ $= \int_0^a f(-x) dx$	2 marks – correct solution 1 mark – obtains $\int_a^0 f(-u)(-du)$

b-ii	$\int_{-a}^a f(x)dx = \int_{-a}^0 f(x)dx + \int_0^a f(x)dx$ $= \int_0^a f(-x)dx + \int_0^a f(x)dx$ $= \int_0^a \{f(-x) + f(x)\}dx$	1 mark – correct solution
b-iii	$\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \frac{1}{1 + \sin x} dx$ $= \int_0^{\frac{\pi}{6}} \left\{ \frac{1}{1 + \sin(-x)} + \frac{1}{1 + \sin x} \right\} dx$ $= \int_0^{\frac{\pi}{6}} \left\{ \frac{1}{1 - \sin x} + \frac{1}{1 + \sin x} \right\} dx$ $= \int_0^{\frac{\pi}{6}} \left\{ \frac{1 + \sin x + 1 - \sin x}{1 - \sin^2 x} \right\} dx$ $= \int_0^{\frac{\pi}{6}} \left\{ \frac{2}{\cos^2 x} \right\} dx$ $= 2 \int_0^{\frac{\pi}{6}} \sec^2 x dx$ $= 2 \left[\tan x \right]_0^{\frac{\pi}{6}}$ $= 2 \left\{ \tan \frac{\pi}{6} - \tan^0 \right\}$ $= 2 \left\{ \frac{1}{\sqrt{3}} - 0 \right\}$ $= \frac{2\sqrt{3}}{3}$	3 marks – correct solution 2 marks – forms a correct expression 1 mark – applies result in b(ii) to $f(x) = \frac{1}{1 + \sin x}$

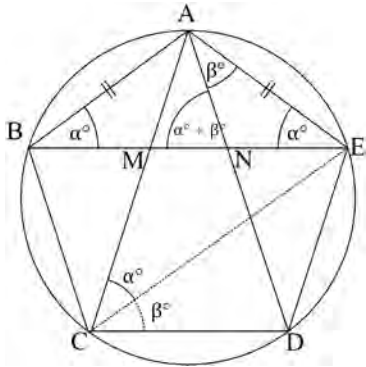
c-i	 $T_x = T \sin \theta$ $T_y = T \cos \theta = W = mg$ $\therefore T = \frac{mg}{\cos \theta} \quad (1)$ $T = F_{c_2}$ $T \sin \theta = \frac{mv^2}{r} \quad (2)$ Substitute for T $\frac{mg}{\cos \theta} \sin \theta = \frac{mv^2}{r}$ $g \tan \theta = \frac{v^2}{r}$ $\tan \theta = \frac{v^2}{rg}$	3 marks – correct solution 2 marks – obtains equations (1) and (2). 1 marks – obtains equations (1) or (2).
c-ii	$T = \frac{mg}{\cos \theta}$ $T = \frac{0.5 \times 9.8}{\cos 30^\circ} = 5.66 \text{ N}$	1 mark – correct solution
c-iii	 $\sin 30^\circ = \frac{r}{1} \Rightarrow r = \frac{1}{2}$ $v^2 = rg \tan \theta$ $v^2 = \frac{1}{2} \times 9.8 \times \tan 30^\circ$ $v = \sqrt{2.82901}$ $v = 1.682 \text{ ms}^{-1}$	1 mark – correct solution

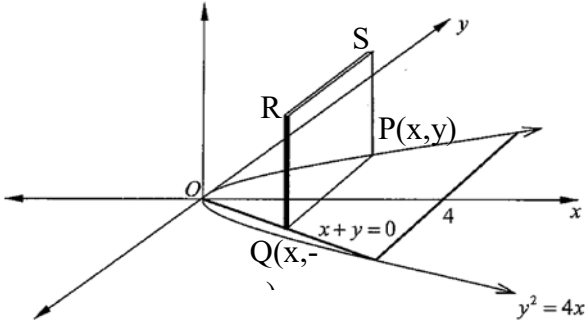
c-iv	$\text{Period} = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{v}{r}} = \frac{2\pi r}{v}$ $= \frac{2\pi \times 0.5}{1.682}$ $= 1.9 \text{ s}$	1 mark – correct solution
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Question 15

a-i	$z^9 - 1 = (z^3)^3 - (1)^3$ $= (z^3 - 1) \left((z^3)^2 + z^3 \times 1 + 1^3 \right)$ $= (z^3 - 1)(z^6 + z^3 + 1)$ $= (z - 1)(z^2 + z + 1)(z^6 + z^3 + 1)$	1 mark – correct solution
a-ii	$\Rightarrow \begin{aligned} a^n - b^n &= (a - b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1}) \\ z^9 - 1 &= (z - 1)(z^8 + z^7 + \dots + 1) \\ (z - 1)(z^8 + z^7 + \dots + 1) &= (z - 1)(z^2 + z + 1)(z^6 + z^3 + 1) \\ (z^8 + z^7 + \dots + 1) &= (z^2 + z + 1)(z^6 + z^3 + 1) \end{aligned}$	1 mark – correct solution

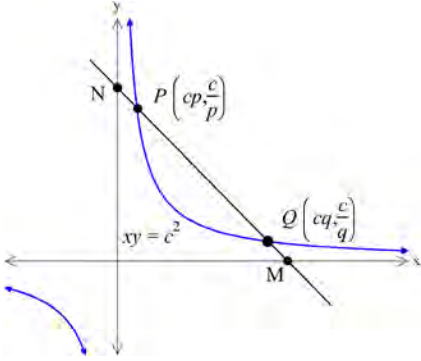
a-iii	$z^9 - 1 = 0 \Rightarrow z^9 = 1$ Let $z = \text{cis}\theta$ $z^9 = (\text{cis}\theta)^9 = \text{cis}(9\theta)$ $\cos(9\theta) + i\sin(9\theta) = 1$ $\therefore \cos(9\theta) = 1$ $9\theta = 2n\pi$ $\theta = \frac{2n\pi}{9} \quad n = \pm(0, 1, 2, 3, \dots)$ $z = \text{cis}\frac{2n\pi}{9}$ $z = 1, \text{cis}\left(\pm\frac{2\pi}{9}\right), \text{cis}\left(\pm\frac{4\pi}{9}\right), \text{cis}\left(\pm\frac{6\pi}{9}\right), \text{cis}\left(\pm\frac{8\pi}{9}\right)$ But solutions of $z^3 - 1 = 0$ $z = 1, \text{cis}\left(\pm\frac{6\pi}{9}\right) = \text{cis}\left(\pm\frac{2\pi}{9}\right)$ $\therefore \text{Solutions of } z^6 + z^3 + 1 = 0$ $z = \text{cis}\left(\pm\frac{2\pi}{9}\right), \text{cis}\left(\pm\frac{4\pi}{9}\right), \text{cis}\left(\pm\frac{8\pi}{9}\right)$	2 marks – correct solution 1 mark – obtains $z = \text{cis}\frac{2\pi}{9}$
a-iv	$z^6 + z^3 + 1 = 0$ $\text{Sum of roots} = -\frac{b}{a} = 0$ $\text{cis}\left(\frac{2\pi}{9}\right) + \text{cis}\left(-\frac{2\pi}{9}\right) + \text{cis}\left(\frac{4\pi}{9}\right) + \text{cis}\left(-\frac{4\pi}{9}\right) + \text{cis}\left(\frac{8\pi}{9}\right) + \text{cis}\left(-\frac{8\pi}{9}\right) = 0 + 0i$ $\cos\left(\frac{2\pi}{9}\right) + \cos\left(-\frac{2\pi}{9}\right) + \cos\left(\frac{4\pi}{9}\right) + \cos\left(-\frac{4\pi}{9}\right) + \cos\left(\frac{8\pi}{9}\right) + \cos\left(-\frac{8\pi}{9}\right) = 0$ nb $\cos\theta = \cos(-\theta)$ $2\cos\left(\frac{2\pi}{9}\right) + 2\cos\left(\frac{4\pi}{9}\right) + 2\cos\left(\frac{8\pi}{9}\right) = 0$ $\cos\left(\frac{2\pi}{9}\right) + \cos\left(\frac{4\pi}{9}\right) + \cos\left(\frac{8\pi}{9}\right) = 0$ $\cos\left(\frac{2\pi}{9}\right) + \cos\left(\frac{4\pi}{9}\right) = -\cos\left(\frac{8\pi}{9}\right)$ $\cos\left(\frac{2\pi}{9}\right) + \cos\left(\frac{4\pi}{9}\right) = -\cos\left(\pi - \frac{\pi}{9}\right)$ $\cos\left(\frac{2\pi}{9}\right) + \cos\left(\frac{4\pi}{9}\right) = -\left\{\cos\pi\cos\frac{\pi}{9} - \sin\pi\sin\frac{\pi}{9}\right\}$ $\cos\left(\frac{2\pi}{9}\right) + \cos\left(\frac{4\pi}{9}\right) = -\left\{-\cos\frac{\pi}{9}\right\}$ $\cos\left(\frac{2\pi}{9}\right) + \cos\left(\frac{4\pi}{9}\right) = \cos\frac{\pi}{9}$	2 marks – correct solution 1 mark – uses sum of roots to form an equation

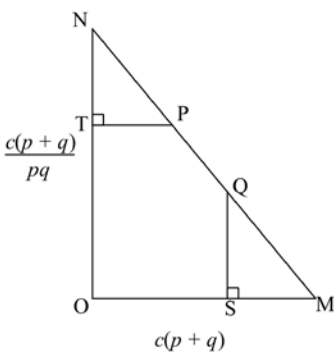
b-i	 Let $\angle BEA = \alpha$ $AB = AE$ given $\therefore \triangle ABE$ is isosceles $\angle ABE = \alpha$ (base angles of isosceles triangle) $\angle ACE = \angle BEA = \alpha$ (= $\angle$'s on chord AE)	2 marks correct solution 1 mark - $\angle ABE = \alpha$ with reason or equivalent.
b-ii	Let $\angle EAN = \beta$ $\angle ECD = \beta$ (equal $\angle$'s on chord ED) $\angle ACD = \alpha + \beta$ $\angle ANM = \alpha + \beta$ (ext $\angle$ of $\triangle$) $= \angle ACD$ $\therefore CDNM$ is cyclic quad (ext $\angle =$ opp interior $\angle$)	3 marks – correct solution 2 marks – correct use of 2 circle geometry rules applicable to question 1 mark - correct use of 1 circle geometry rule applicable to question

c-i	 $PQ = x + y$ $\therefore SP = x + y$ $\text{Area} = (y + x)^2$ $= y^2 + 2xy + x^2$ $\text{But } y^2 = 4x$ $y = \sqrt{4x} = 2\sqrt{x}$ $\therefore A = 4x + 2x \cdot 2\sqrt{x} + x^2$ $= 4x + x^2 + 4x^{\frac{3}{2}}$	2 marks – correct solution 1 mark – forms $PQ \text{ (or } SP) = x + y$
c-ii	$\delta V = \text{area} \times \delta x$ $V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^4 \left\{ 4x + x^2 + 4x^{\frac{3}{2}} \right\} \delta x$ $= \int_0^4 \left\{ 4x + x^2 + 4x^{\frac{3}{2}} \right\} dx$ $V = \left[2x^2 + \frac{x^3}{3} + \frac{8x^{\frac{5}{2}}}{5} \right]_0^4$ $= \frac{1568}{15}$	2 marks – correct solution including development of integral 1 mark – correct primitive function or correct evaluation from (equivalent) incorrect primitive.

Question 16

a-i	$I_n = \int_0^1 (1-x^2)^n dx$ $u = (1-x^2)^n \quad dv = 1$ $du = -2nx(1-x^2)^{n-1} \quad v = x$ $I_n = \left[x(1-x^2)^n \right]_{(0,1)} - \int_0^1 -2nx^2(1-x^2)^{n-1} dx$ $= 0 - 2n \int_0^1 -x^2(1-x^2)^{n-1} dx$ $= -2n \int_0^1 \left\{ (1-x^2) - 1 \right\} (1-x^2)^{n-1} dx$ $= -2n \int_0^1 (1-x^2)^n - (1-x^2)^{n-1} dx$ $= -2n \int_0^1 (1-x^2)^n dx + 2n \int_0^1 (1-x^2)^{n-1} dx$ $I_n = -2nI_n + 2nI_{n-1}$ $2nI_n + I_n = 2nI_{n-1}$ $(2n+1)I_n = 2nI_{n-1}$ $I_n = \frac{2nI_{n-1}}{2n+1}$	3 marks – correct solution 2 marks – rewrites initial expression for I_n in terms of integrals for I_n and I_{n-1} 1 mark – applies IBP to obtain integral expression to I_n and I_{n-1}
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a-ii	$I_3 = \frac{6}{7} I_2 = \frac{6}{7} \times \frac{4}{5} I_1$ $I_1 = \int_0^1 (1-x^2) dx$ $= \left[x - \frac{x^3}{3} \right]_0^1 = \frac{2}{3}$ $I_3 = \frac{6}{7} \times \frac{4}{5} \times \frac{2}{3} = \frac{16}{35}$	1 mark – correct solution
b-i	 $m_{PQ} = \frac{\frac{c}{p} - \frac{c}{q}}{cp - cq}$ $= \frac{c \left(\frac{q-p}{pq} \right)}{c(p-q)}$ $= -\frac{1}{pq}$ $y - y_1 = -\frac{1}{pq}(x - x_1)$ $y - \frac{c}{p} = -\frac{1}{pq}(x - cp)$ $pqy - cp = -x + cp$ $x + pqy = c(p+q)$	2 marks – correct solution 1 mark – finds correct simplified gradient.

b-ii	 At $M, y = 0 \Rightarrow x = c(p + q)$ At $N, x = 0 \Rightarrow y = \frac{c(p + q)}{pq}$ $\therefore y = \frac{cp}{pq} + \frac{cq}{pq}$ $= \frac{c}{q} + \frac{c}{p}$ $\therefore NR = \frac{c}{q} + \frac{c}{p} - \frac{c}{p} = \frac{c}{q}$ In $\triangle NRP, \triangle QSM$ $\angle RNP = \angle SQM$ (corr $\angle$'s on $\parallel$ lines) $\angle NRP = \angle QSM = 90^\circ$ given/ construct $NR = QS = \frac{c}{q}$ $\triangle NRP = \triangle QSM$ (AAS) $PN \equiv QM$ (corresp sides in cong Δ)	3 marks – correct answer 2 marks – finds NR 1 mark – some
c-i	$P(x) = (x^2 - a^2)Q(x) + px + p$ $= (x - a)(x + a)Q(x) + px + p$ $\therefore P(a) = pa + p \quad (1)$ $P(-a) = -pa + p \quad (2)$ $(1) - (2)$ $P(a) - P(-a) = pa - (-pa) = 2pa$ $p = \frac{1}{2a} \{P(a) - P(-a)\}$ $(1) + (2)$ $P(a) + P(-a) = pa + (-pa) = 2p$ $q = \frac{1}{2} \{P(a) + P(-a)\}$	3 marks – correct solution 2 marks – obtains correct expressions for $P(a)$ and $P(-a)$. 1 mark – obtains a correct expression for either $P(a)$ or $P(-a)$.

c-ii	$P(x) = x^n - a^n$ If n is even $P(a) = a^n - a^n = 0$ $P(-a) = (-a)^n - a^n = a^n - a^n = 0$ $\therefore$ from part i $p = 0, q = 0$ $\therefore$ no remainder If n is odd $P(a) = a^n - a^n = 0$ $P(-a) = (-a)^n - a^n = -a^n - a^n = -2a^n$ $\Rightarrow p = \frac{1}{2a} \{0 - (-2a^n)\} = a^{n-1}$ $q = \frac{1}{2} \{0 + (-2a^n)\} = -a^n$ $R(x) = px + q = a^{n-1}x - a^n$	3 marks – correct solution 2 marks – distinguishes two cases and solves one case correctly. 1 mark – correctly considers 1 case only.
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